

SANTIAGO MOLINER PREJANO

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Software engineer specializing in modern C++, real-time simulation, distributed systems, and synthetic environment generation. Experienced in aerospace and defense programs, developing high-performance simulation frameworks, image generation pipelines, and scalable infrastructure for autonomy and machine learning applications.

Professional Experience

AIRBUS Defence & Space

Madrid, Spain Feb 2025 – Present

Simulation Engineer

- Developed high-performance deterministic real-time simulation models for the SIRTAP program using modern C++20, validated with ASAN and UBSAN to ensure runtime reliability and memory safety.
- Contributed to the open-source SEN framework by developing low-latency real-time visualization, telemetry, and monitoring tools.
- Designed and implemented a code-generation framework that produces C++ communication layers from data definitions, automating interoperability between disparate middleware and communication frameworks.
- Designed and developed an Unreal Engine-based synthetic data generation system integrated with SEN for ML training in MRTT automated air-to-air refueling and SIRTAP automated landing, generating pixel-accurate ground-truth data, synchronized sensor outputs, and segmentation masks for supervised learning workflows.
- Led cross-project technical coordination to define and standardize interfaces across distributed, service-oriented architectures.
- Architected and deployed a Kubernetes-based platform for a centralized geospatial data service, enabling scalable access for simulation, visualization, and synthetic image-generation systems.

Bertrandt (AIRBUS Subcontractor)

Madrid, Spain Sep 2023 – Feb 2025

Simulation Engineer

- Developed the real-time sensor model simulation for gimbal control and entity tracking within the SERPAS project, integrating it with other simulation models.
- Integrated and validated multiple CIGI-compliant Image Generators, including GenesisIG (Diamond Visionics), VBS Blue IG (Bohemia Interactive), and Aechelon.
- Designed comprehensive unit tests for the sensor model simulation to ensure reliability and performance, validating both accuracy and seamless integration with real-time data streams.
- Developed a gateway for the Airbus software communication framework (SEN), enabling translation between simulation protocols such as DIS and CIGI with minimal latency.

Fortra

Barcelona, Spain Nov 2021 – Sep 2023

Cybersecurity Software Engineer

- Developed and maintained a custom penetration testing tool (Core Impact) primarily in C++17 and Python.
- Maintained and enhanced the open-source Impacket library, managing pull requests and adding new features.
- Developed a Python-based NTLM relay attack with Impacket to intercept and relay NTLMv2 hashes over SMB, integrating Mimikatz for efficient offline password extraction within Active Directory environments.
- Performed penetration testing and vulnerability assessments on Windows-based environments using Kali Linux and industry-standard offensive security tools.

Education

Universidad Politécnica de Cataluña (UPC)

Barcelona, Spain 2019 - 2022

Bachelor's Degree in Video Game Development

- **Specializations:** Advanced graphics programming, AR/VR, Networking

Technical Skills

Programming Languages: C++20, C, x86/x64 Assembly, Python, Zig, Rust

Technologies & Tools: Linux, Git, C/C++ Sanitizers, Docker, Kubernetes, CMake, Conan, Unreal Engine, Godot